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PALEOMAGNETISM OF SAN CRISTOBAL ISLAND, GALAPAGOS *

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Isla San Cristobal, the most easterly of the Galapagos Islands, consists of two parts: a large volcano constitutes the southwest half of the island and an irregular apron of small cones and flows makes up the northeast half. As some of the younger flows on the flanks of the large volcano are reversely magnetized, the minimum age of the volcano is 0.7 my, which is the age of the Brunhes–Matuyama reversal boundary. The true age is probably several times greater. The cones and flows to the northeast are all normally magnetized. The between-site angular dispersion of virtual poles is 11.3° – a value consistent with mathematical models for the latitude dependence of geomagnetic secular variation.

1. Introduction

The Galapagos Islands, because of their location on the equator, provide a particularly interesting record of geomagnetic secular variation. Moreover, the age of the islands is of interest from the viewpoint of plate tectonics because of their geographic position. The islands are located about 1000 km off the coast of Ecuador on the convex side of a pronounced bend in the East Pacific Rise, which lies about 1000 km to the west of the islands. Typical oceanic magnetic lineations have been identified in the vicinity of the Galapagos Islands, but surprisingly, these trend east-west (fig. 1) rather than north-south, as they would had they been produced along the East Pacific Rise. This has led Herron and Heirtzler [1] and Raff [2] to propose that an east-west rift zone exists at 0.8° north latitude. Because of the proximity of the Galapagos Islands to this rift zone, their age is of particular interest as the maximum age of a rather

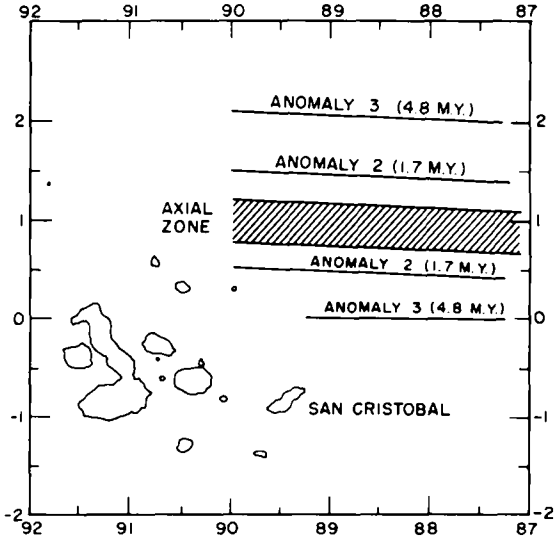


Fig. 1. Index map showing the Galapagos Archipelago. The axial zone (shaded) along which new ocean floor is forming is from Herron and Heirtzler [1] and Raff [2]. The locations of magnetic anomalies 2 and 3 are from Herron and Heirtzler [1]; the ages shown for the anomalies are from Heirtzler et al. [3] and Cox [4].

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young part of the sea floor. The magnetic anomaly pattern has not been surveyed in the immediate vicinity of the islands, but east–west lineations have been observed both to the northeast and southeast of the Galapagos. Extrapolating from the pattern to the north, the expected age of the sea floor in the vicinity of San Cristobal is 9 to 10 my, which is the maximum age of the islands consistent with the existence of the proposed rift zones.

Cox and Dalrymple [5] earlier described the paleomagnetic stratigraphy and gave radiometric ages of most of the islands but not of San Cristobal. They found that the large islands in the western part of the archipelago were made up of young, normally magnetized lava flows that had formed during the Brunhes normal polarity epoch (0 to 0.7 my old). Older rocks, including reversely magnetized lava flows that had formed during the Matuyama reversed polarity epoch (0.7 to 2.4 my), were found on the more easterly islands. The present study extends the paleomagnetic stratigraphy of the Galapagos to San Cristobal, the most easterly of the islands.

2. Sampling

The objective of the paleomagnetic sampling was to obtain samples spanning as much of the history of the island as possible. Some of the basalt and palagonite cones on the island were described briefly by Darwin [7] and by McBirney and Williams [8], but the island has not been mapped geologically. Therefore, in the present study it was necessary to rely on available topographic maps and air photos in planning the sampling. Topographically San Cristobal consists of two distinct parts. The southwest part of the island is dominated by a single large volcano with steep flanks and a crater lake at its top (fig. 2). The northeastern half of the island is an apron built up of irregular lava flows and small cones. The northeast part of the island is clearly the younger. Some of the youngest flows are nearly barren of vegetation and have undergone no erosion by the surf where they enter the sea. The large volcano to the southwest is densely vegetated and has undergone considerable erosion, especially on its south flank. Because of surf conditions which made

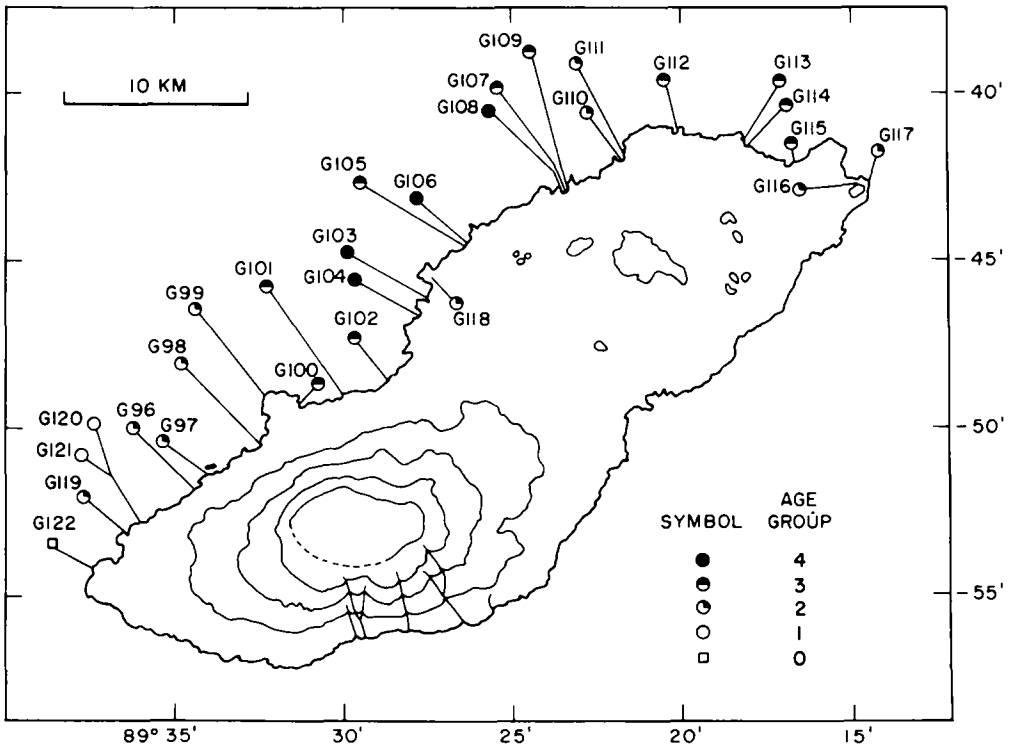


Fig. 2. San Cristobal Island. Sample localities are indicated by lines from the age group symbols. Age group 0 is the oldest and 4 the youngest. Contours are from [6] and are at intervals of 500 feet.

it impossible to land on the southern coast of the island, sampling was confined to the 50-km-long north coast. The sites range in age from the youngest lava flows on the northeast half of the island to some of the older flows on the flanks of the large volcano, with a good distribution of sites of intermediate age.

Sets of oriented samples were collected at sites in 27 lava flows. The site locations are specified to within 100 m by the latitude and longitude coordinates in table 1, which are accurate to 0.001°, as read from standard charts [6]. Each site was assigned to one of the following five age groups on the basis of the extent of erosion and amount of vegetation, as determined from air photo and field studies.

Age group 4. The coastline where the flow has entered the sea is irregular. Erosion by the surf has been minor, removing only loose debris and, at most, 0.5 m of the flow surface. The color of the flow surface is dark grey or black. There is no vegetation on the flow other than a few solitary small cactus plants on the flow surface and at most a few mangrove trees along the coast.

Age group 3. The front of the flow where it entered the sea has been moderately trimmed by the sea to form a small scarp 0.5 m to 2 m high. Color of rock surface is brown. There is scattered to moderately dense vegetation, but 20% or more of the original flow surface appears as bare rock on the air photos. Original flow surfaces are generally preserved beneath the vegetation, and there is no indication of stream erosion or soil formation.

Age group 2. Along the coast, the original irregular outline of the flow has been trimmed by the surf to form a rather smooth coastline. Small sea cliffs may have been formed, and small cones near the coast are truncated. The flow may be faulted. On the surface of the lava, original flow features are largely obliterated by erosion and there may be small stream channels. Soil may be developed on the flow surface, and the vegetation is dense. Magnetic polarity is normal.

Age group 1. Similar to 2, but the magnetic polarity is reversed; the flow belongs to the Matuyama reversed polarity epoch (2.4 to 0.7 my ago).

Age group 0. Similar to 1 and 2, but on the basis of stratigraphy, the flow is probably older than the Matuyama reversed polarity epoch.

The above criteria of age are in no sense absolute. For example, the surf runs strongest from the south; hence surf erosion proceeds more slowly along north coasts, especially in sheltered bays. Rainfall is greater at higher elevations; therefore rates of surface erosion, rates of soil formation, and rates of vegetation growth are also greater. All the sites for the present study were along the arid coast, where the rates of surface erosion, soil formation, and plant growth are extremely slow – so slow, in fact, that flows with ages of 0.7 my are distinctly more youthful in appearance than the older flows on the island. Using these criteria, flows younger than 0.7 my can be placed in a relative stratigraphic sequence with considerable confidence.

All but two of the 27 sampling sites (fig. 2) were in or near the intertidal zone . Most of the cones were drilled where the surf had removed loose blocks or had stripped the tops of the flows to expose their massive interiors. Almost none of the magnetization of the samples appears to have been changed by the rotation of blocks of lava during cooling or by lightning strikes near the sample sites. As a result, the intrasite angular dispersion in the magnetic directions is unusually small.

In order to be able to measure changes in ancient field direction with an accuracy of several degrees, eight samples were collected at each site to permit reduction of experimental errors by averaging. Cores were cut using a diamond drill and were oriented with respect to geographic north. The overall accuracy of core orientation was about 2°.

3. Magnetic results

Measurements of remanent magnetization were made using a standard spinner magnetometer with a precision of 1° (angular measurements) and 2% (intensity measurements) (table 1). To assess magnetic stability, specimens from 12 sites were demagnetized in alternating magnetic fields of 12.5, 25, 50, 100, 200, 400, 600, and 800 Oe. On average, a field of 153 Oe was required to reduce the remanence to half its original value, and an alternating field of about 600 Oe was required to reduce the remanence by 90%. Such high coercive forces are typical of the thermoremanence acquired by basalt when it cools in the earth's magnetic field.

At sites with moderate amounts of scatter, it was

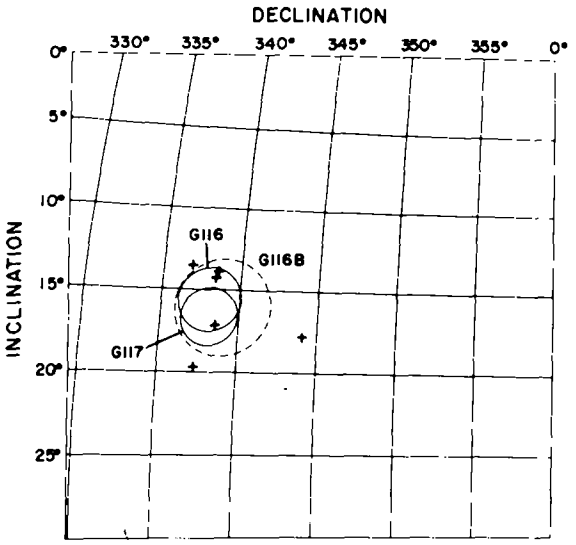


Fig. 3. Results from sites G116 and G117 in lava flows from the same small cone and from site G116B, a zone of baked sediments beneath the flow at site G116. Solid circles: 95% confidence cones for sites G116 and G117; crosses: directions of magnetization for individual samples from site G116B; dashed circle: 95% confidence cone for site G116B.

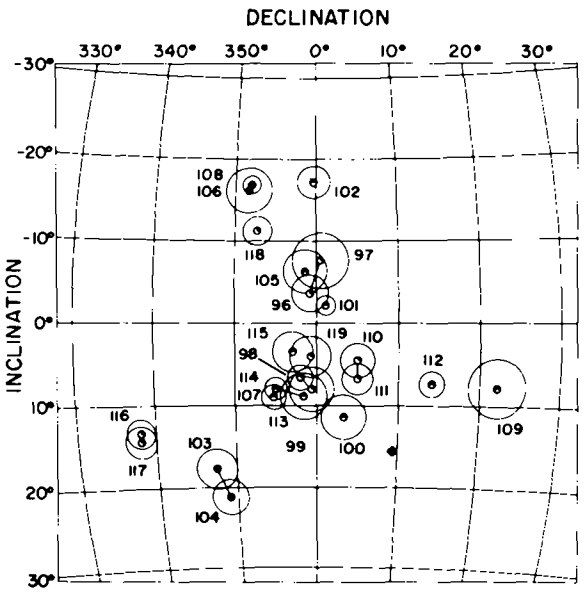


Fig. 4. Average directions of magnetization for all sites of the Brunhes normal polarity epoch. Straight lines join directions of sites from lava flows presumed to have the same or nearly the same ages. Large circles: 95% confidence cones around average directions; small circles: average directions of magnetizations, the symbols being as in fig. 2; diamond: present direction of the earth's field.

found that magnetic cleaning in alternating fields of either 50 or 100 Oe was sufficient to raise the value of the precision parameter to 380 or higher. The final values for the 95% confidence cones ranged from 1.1 to 3.6°, with an average of 2.2° (table 1).

An independent check on the accuracy with which the lava flows have recorded ancient magnetic field directions was obtained at sites G116, G116B, and G117 (fig. 3). The first of these sites is from a lava flow in the crater of a small cone. The second (G116B, dashed circle) is from a red zone of sediments baked by this flow. Site G117 is from a lava flow 0.7 km to the north which appears to have erupted from the same cone. All three sites probably became magnetized within a period of a few years or less, as all of them have the same magnetic direction within 5% statistical uncertainty. For site G116 the radius of the 95 percent confidence is 1.8° and for site G117 it is 1.9°.

4. Geomagnetic secular variation

The average directions of magnetization of the 24 sites in the Brunhes normal epoch are plotted individually in fig. 4 and the overall average direction is given in table 2. The magnetic directions are not randomly distributed, as many of them show a distinct tendency to occur in pairs. This suggests that these pairs of sites may be from the same flow or from flows that formed in rapid succession. If this is true, the number of independent observations of ancient field directions is less than 24 and the true confidence limits are broader than indicated in column 1 of table 2. On careful examination of the air photos and field notes it was found that sites G106 and G108, which are adjacent to each other, are undoubtedly from one flow, as are sites G116 and G117 and sites G96 and G97. Four other pairs of sites, G103 and G104, G110 and G111, G113 and G114, and G98 and G99 are also adjacent to each other and have parallel magnetic directions. Although the geologic information suggests that the two sites from each pair are from flows with slightly

Table 1
Site mean magnetic results, listed in order of increasing age.

Site no.	Age group	Site location		Exp.	N	K	α_{95}	Direction		Virtual pole		Intensity × 1000	
		Lat.	Long.					Incl.	Decl.	Lat.	Long.	Mean value	SD
G-106	4	-0.743	270.561	NRM	8	404	2.8	-16.2	351.7	78.8	138.1	5.60	2.87
G-108 *	4	-0.717	270.609	NRM	8	2586	1.1	-16.8	352.0	78.8	135.7	5.38	1.42
G-103	4	-0.768	270.541	NRM	8	514	2.4	17.7	347.5	74.1	219.1	2.56	0.64
G-104	4	-0.778	270.537	NRM	8	207	4.2	18.7	347.0	73.4	219.8	1.84	0.73
G-104 *	4	-0.778	270.537	050	8	655	2.2	21.2	349.0	74.0	227.8	1.48	0.52
G-109 *	3	-0.715	270.610	NRM	8	244	3.6	7.8	022.2	67.4	348.6	1.27	0.64
G-105 *	3	-0.744	270.56	NRM	8	437	2.6	-6.3	358.7	87.2	119.5	1.76	0.64
G-102	3	-0.810	270.522	NRM	8	109	5.3	-11.5	358.7	84.9	105.1	1.63	0.24
G-102 *	3	-0.810	270.522	100	8	792	2.0	-17.2	359.7	82.0	092.6	1.04	0.20
G-112	3	-0.686	270.665	NRM	8	341	3.3	5.9	012.2	77.2	344.0	2.41	0.89
G-112 *	3	-0.686	270.665	100	8	1260	1.6	7.4	014.1	75.2	343.2	1.98	1.10
G-101 *	3	0.819	270.499	NRM	8	2194	1.2	-2.2	001.1	88.8	013.3	4.58	2.93
G-100	3	-0.822	270.478	NRM	8	267	3.4	10.2	003.8	82.9	302.6	1.78	0.22
G-100 *	3	-0.822	270.478	100	8	416	2.7	11.4	003.3	82.7	296.8	1.57	0.18
G-107	3	-0.717	270.610	NRM	8	303	3.2	7.0	356.5	84.5	231.2	1.41	1.61
G-107 *	3	-0.717	270.610	050	8	1379	1.5	9.0	354.8	82.6	225.7	1.33	1.60
G-114 *	3	-0.693	270.700	NRM	8	1668	1.4	7.9	355.0	83.2	224.0	2.13	0.22
G-113	3	-0.690	270.697	NRM	8	123	5.0	6.2	001.8	85.8	295.3	2.26	0.55
G-113	3	-0.690	270.697	100	8	387	2.8	8.8	358.4	84.6	253.6	1.35	0.48
G-115 *	3	0.702	270.723	NRM	8	512	2.4	3.4	357.1	86.2	220.3	1.62	0.93

Table 1 (continued)

G-110	2	- 0.700	270.639	NRM	8	35	9.5	4.3	002.5	86.2	312.1	1.35	0.56
G-110	2	- 0.700	270.639	100	8	703	2.1	4.5	005.0	84.2	329.9	0.69	0.30
G-111	2	- 0.696	270.639	NRM	8	61	7.2	1.5	002.0	87.5	324.0	1.07	0.11
G-111 *	2	- 0.696	270.639	050	8	871	1.9	6.8	005.0	83.6	321.1	0.81	0.13
G-116	2	- 0.712	270.753	NRM	6	892	2.2	13.2	338.5	67.3	200.1	10.3	3.3
G-116*	2	- 0.712	270.753	050	8	996	1.8	13.4	338.1	66.9	200.0	8.59	3.98
G-116 B	2	- 0.712	270.753	050	6	514	3.0	13.9	338.9	67.6	201.4	0.91	0.26
G-117	2	- 7.111	270.759	NRM	8	816	1.9	14.4	338.1	66.7	201.3	2.81	0.62
G-118	2	- 0.759	270.543	NRM	7	1248	1.7	- 11.4	354.1	82.3	140.1	1.93	0.97
G-118 *	2	- 0.759	270.543	100	8	993	1.8	- 11.2	352.7	81.2	146.3	1.76	0.85
G-98	2	- 0.841	270.458	NRM	8	438	2.6	6.6	357.3	85.1	237.6	11.0	4.1
G-98 *	2	- 0.841	270.458	050	8	1295	1.5	6.6	358.0	85.4	224.8	12.2	4.3
G-99	2	- 0.816	270.460	NRM	8	422	2.7	8.0	359.4	85.1	263.4	2.39	0.52
G-96	2	- 0.867	270.423	NRM	8	140	4.7	- 1.4	358.2	88.2	185.2	3.75	0.68
G-96 *	2	- 0.867	270.423	100	8	617	2.2	- 3.7	359.2	88.8	129.6	2.62	1.00
G-97	2	- 0.856	270.433	NRM	8	264	3.4	- 7.6	000.5	87.0	080.4	4.03	1.06
G-119	2	- 0.886	270.393	NRM	8	487	2.5	4.0	359.2	87.0	255.3	3.99	0.72
G-120	1	- 0.881	270.402	NRM	8	451	2.6	5.2	166.8	- 76.6	352.7	2.70	1.12
G-121	1	- 0.881	270.402	NRM	8	838	1.9	- 11.2	161.8	70.7	020.3	4.34	1.91
G-122	0	- 0.903	270.377	NRM	8	135	4.8	8.2	004.2	83.4	310.6	1.77	0.58
G-122	0	- 0.903	270.377	050	8	744	2.0	9.6	001.9	84.0	288.7	1.76	0.63

Brackets include sites that are correlative.
Site : data with * were used for final statistical analysis.
N : number of oriented samples. At site 118, one aberrant sample was excluded from the NRM analysis.
Experiment : NRM original magnetization. Numbers, peak alternating field used for cleaning.
K : Fisher precision parameter.
α_{95} : 95% confidence limits.
Virtual poles : Latitude and east longitude of negative magnetic pole.
Intensity : Units of 10^{-3} gauss (i.e., emu/cm^3).

Table 2
Statistical analysis of magnetic directions.

	All sites	Nonredundant sites
Number of sites	24	17
Average inclination	3.0°	2.5°
Average declination	357.6°	358.8°
One-level analysis:		
R (sum of site unit vectors)	23.2845	16.5088
Precision parameter K	32	33
α_{95}	5.3°	6.4°
Two-level analysis:		
Within-site K	601	705
Between-site K	32	33
Between-site angular standard deviation s_B	14.2°	14.2°
s_B corrected to axial dipole direction	15.2°	14.9°
95% confidence limits on s_B		12.0°–19.6°
α_{95}	5.0°	6.0°

Table 3
Statistical analysis of virtual geomagnetic poles.

	All sites	Nonredundant sites
Number of sites	24	17
Average VGP latitude	86.6°	87.6°
Average VGP longitude	223.1°	239.3°
One-level analysis:		
R (sum of site unit vectors)	23.5801	16.7063
Precision parameter K	55	54
α_{95}	4.0°	4.9°
Two-level analysis:		
Within-site K	662	806
Between-site K	55	55
Between-site angular standard deviation s_B	10.9°	10.9°
s_B corrected to North Pole	11.5°	11.3°
95% confidence limits on s_B		9.1°–14.8°
α_{95}	3.9°	4.6°

different ages, the magnetic data and the proximity of the flows to each other suggest that in each case the two flows formed in rapid succession. For statistical purposes the number of independent observations is therefore 17 rather than 24. In the second column of table 2 the statistical averages are given, omitting the seven redundant observations and retaining the more accurate of the two measurements in each pair. The results of the statistical analysis of the 17 independent data appear to be realistic. The confidence limit for the overall mean is 6.0° and the mean direction is not significantly different from the direction of an axial dipole field.

The angular dispersion in field directions due to geomagnetic secular variation is given in table 2, where a two-level analysis [9] was done to eliminate angular dispersion due to experimental noise (within-site K). A virtual geomagnetic pole was then calculated for each site and a similar statistical analysis of poles was performed (table 3). The value of the precision parameter K for the wobble of the poles is 55, corresponding to an angular standard deviation of 10.9°. Comparing the values of K found in the one-level and two-level analyses, it can be seen that the effect of experimental noise on the within-site level is negligible. A small correction of the wobble from 10.9° to 11.3° is appropriate if it is assumed, as is probably correct, that the true mean pole is the north geographic pole.

The 95% confidence interval for the angular standard deviation of virtual poles [10] is from 9.1° to 14.8°. All the models [11–15] that have been advanced to describe the latitude dependence of secular variation fall within this range, which is broad because the number of independent observations is limited. The present result supports the general conclusion of these models that the wobble of virtual pole, as seen from an equatorial sampling area, should be small. Similar studies are in progress on other islands of the Galapagos. It is anticipated that the results of these individual studies will provide a useful test of the statistical methods here employed and that the final combined data set of independent determinations from all the islands will permit determination of the angular wobble with sufficient accuracy to test models for secular variation.

5. Age of lavas

The youngest looking flow in age group 4, that at

site G106, appears to be no more than a few decades old. There is no vegetation on its upper surface other than a few solitary cactus plants, and although the surf is very active, it has done little other than remove blocks of loose rubble from the surface of the flow. Therefore it would not be surprising if the direction of magnetization were close to that of the present field. Yet it differs from the present field by 36°. This angular difference cannot be translated into an age difference because the geomagnetic field does not change direction at a constant rate. Yet the rate, though highly variable, appears to have an upper limit of 20° per century. Moreover, in the central part of the Pacific region, the direction of the magnetic field appears generally to change at a rate of a few degrees per century or less [16]. The flow at site G106 is therefore at least several centuries old; it is probably considerably older. The flow at site G108, 6 km to the northeast, has the same magnetic direction and appears on the air photos to be part of the same broad flow; although mangrove trees were noted along the front of G108 and on the ground, it appears slightly older than G106. The magnetic results indicate that the two are the same. Similarly sites G103 and G104 in age group 4 are correlative with each other but are clearly older than G108 and G106. Of interest is the fact that during the time represented by this youngest age group, probably only a small fraction of the Brunhes normal polarity epoch, the amplitude of angular variations is nearly as great as it was during the entire 700 000 years of the epoch.

The amount of erosion and the extent of vegetation on the flows is progressively greater in age groups 3 and 2. In selecting the criteria which define these two groups, an attempt was made to divide the Brunhes normal polarity epoch into two parts of approximately equal length, assuming that rates of erosion and of plant colonization on San Cristobal were similar to rates on other islands of the archipelago with known radiometric ages [5].

This set of qualitative relative ages on San Cristobal (fig. 5) is tied to an absolute time scale at only one point, 0.7 my. This is the age of the boundary between the Matuyama and Brunhes polarity epochs, which is placed between age groups 1 and 2 on the following basis. The flows of the two groups have similar amounts of vegetation and have undergone similar

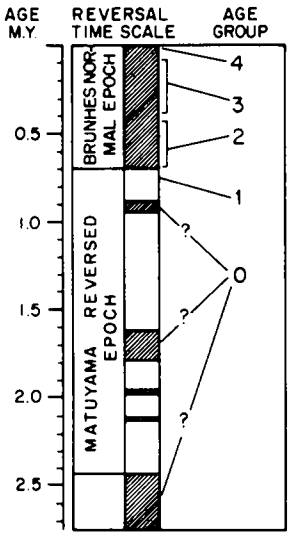


Fig. 5. Correlation of age groups to the reversal time scale [4].

amounts of erosion. Moreover, there is no indication of a hiatus in volcanic activity between them. This is especially evident at site G119 (normal) and sites G120 and G121 (reversed), where flows of the two groups are closely juxtaposed. Site G119 is a normally magnetized flow from a small volcanic cone (shown as Cerro Chivo on [6]) which has been partly eroded to expose agglomerate layers and ancient beach deposits. Sites G120 and G121 are in two superposed reversely magnetized lavas from another small volcanic vent located only 1.2 km northeast of G119. Here too the surf has cut the cone to expose agglomerate layers and beach deposits similar to those at Cerro Chivo. On the air photos the terrain in the vicinity of the two cones has similar amounts of vegetation and the amounts of erosion and faulting are similar. Probably these lava flows formed slightly before and slightly after one or more of the three polarity reversals that occurred between 1.0 and 0.7 my ago (fig. 5). The progressively younger flows in age group 2 probably formed during the early part of Brunhes normal epoch.

The lava sampled at site G122 is more heavily vegetated than any of the other flows and appears older, but whether it formed prior to the Matuyama reversed epoch or during one of the normal events within the Matuyama epoch is uncertain. Whichever is correct, the magnetic stratigraphy establishes that the lavas along the north coast of San Cristobal have a maxi-

mum age of about 1 my with the possible exception of the lavas at the extreme southwest end of the coast, which may be older.

Do the basalts that were sampled include the oldest rocks exposed on the islands? From an examination of air photos of the island, the answer is almost certainly no. Sites G119–G122 are in small parasitic cones which mark the final stages of activity of the large central volcano. On the south flank of this same volcano, which was not sampled, the vegetation is more dense and the erosion more advanced. Several streams have cut well-developed channels – a rare occurrence on any of the Galapagos Islands. By comparing this terrain with that at sites G119–G121, where the age is known to be 1.0 to 0.7 my, I estimate that the rocks on the south flank of the central volcano are at least several million years old and that they may be considerably older. This minimum age is consistent with the age of about 10 my for the sea floor in this vicinity, as inferred from the magnetic anomalies to the north.

The oldest rocks formed during the beginning of the volcanic activity that produced Isla San Cristobal are probably either below sea level or buried beneath the outer strata of the volcano. Therefore it would be uncertain, even after complete sampling of surface rocks, whether the volcanism that produced San Cristobal began at the same time that new sea floor was being formed along the ancient ridge. On the other hand, there is no question that volcanism has occurred repeatedly, and possibly continuously, during the 10 my that have elapsed since the ridge migrated north from this area.

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